In the first two problems, we will finally prove the Division Algorithm for non-negative integers.

Problem 1. For $a \in \mathbb{N} \cup \{0\}$, let $P(a)$ be the statement "for all $b \in \mathbb{N}$, there exist $q, r \in \mathbb{N} \cup \{0\}$ such that $a = bq + r$ and $0 \leqslant r < b$."

Prove by strong induction that $P(a)$ is true for all $a \geqslant 0$.

Note. The statement $P(a)$ is a universally quantified statement for b.

Hint. In the inductive step, consider the cases $k + 1 < b$ and $k + 1 \geqslant b$.

Proof. Base case: Our base case is $a = 0$. We need to show that, for every $b \in \mathbb{N}$, there exist $q, r \in \mathbb{N} \cup \{0\}$ such that $0 = bq + r$ and $0 \leqslant r < b$. Then $a = 0 = 0b + 0$, so by setting $q = r = 0$, we are done.

Induction step: Assume that, for some $k \in \mathbb{N} \cup \{0\}$, the statement $P(a)$ holds when $a = 0, 1, \ldots, k$ (i.e., we assume that for every $0 \neq b \in \mathbb{N}$, there exist some $q, r \in \mathbb{N}$ such that $a = bq + r$ and $0 \leqslant r < b$).

We need to show that for $k + 1$, for every $b \in \mathbb{N}$, there exists $q, r \in \mathbb{N}$ such that $k + 1 = bq + r$ with $0 \leqslant r < b$. Since we need to prove this for every such b, let $b \in \mathbb{N}$ be arbitrary.

If $k + 1 < b$, then observe that $k + 1 = 0b + (k + 1)$, so $q = 0$ and $r = k + 1$ meet the criteria.

If instead $k+1 \geqslant b$, then we consider $k+1-b$. We note in this case that $a = k+1-b$ satisfies and $0 \leqslant a \leqslant k$. Applying our induction hypothesis, for this $a = k+1-b$ we can find some $q, r \in \mathbb{N}$ such that $k+1-b = bq+r$ and $0 \leqslant r < b$. Therefore $k+1-b = k+1-b+b = bq+r+b = b(q+1)+r$ where $0 \leqslant r < b$. Thus the statement holds for $k + 1$ as well.

Therefore by induction, this completes the proof. □

Problem 2. Show that the q and r in Problem 1 are unique. That is, show if there also exist q', r' such that $a = bq' + r'$ and $0 \leqslant r' < b$, then $q' = q$ and $r' = r$. You may follow this guide:

1. Prove that $r - r' = b(q' - q)$.

   Proof. Since $a = bq+r$ and $a = bq'+r'$, we obtain $r = a-bq$ and $r' = a-bq'$. Substracting the second equality from the first one, we see that $r-r' = a-bq-(a-bq') = bq'-bq = b(q'-q)$, as desired. □

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2. Show that $-b < r - r' < b$. **Proof.** Since $r' < b$, we have $-b < -r'$. Now, since $0 \leqslant r$, adding this inequalities we obtain $-b < r - r'$. Similarly, using that $0 \leqslant r'$, we have $-r' \leqslant 0$. Since $r < b$, adding this last two we obtain $r-r' < b$. Thus we have proved that $-b < r-r' < b$. □

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3. Use that if x is an integer multiple of b and $-b < x < b$, then $x = 0$. **Proof.** Since $r - r' = b(q' - q)$, we have $-b < b(q' - q) < b$. Since $q, q' \in \mathbb{Z}$, we have $q' - q \in \mathbb{Z}$ and so $b(q' - q)$ is some integer multiple of b. But the only integer multiple of b in this range is 0, so $b(q'-q) = 0$. Since $b \neq 0$, we have $q' - q = 0$. In other words, $q = q'$.

   Now, by part 1, we also have $r - r' = 0$, so $r = r'$. Therefore the q and r really are unique. □

Problem 3. Are the following propositions true or false? Justify your conclusion with proof or counterexample. Throughout, suppose that A, B, C, and D are subsets of a universal set U.

1. $A \subseteq B$ if and only if $B^c \subseteq A^c$.

   Solution. This statement is true.

   Proof. [⇒]. Suppose that $A \subseteq B$. We must show that $B^c \subseteq A^c$. To that end, consider any $x \in B^c$. Then $x \notin B$. Since $A \subseteq B$, we know that if $y \in A$ then $y \in B$. The contrapositive of this

statement is that if $y \notin B$ then $y \notin A$. Therefore, since $x \notin B$, we have $x \notin A$. By the definition of complement, $x \in A^c$. Hence, we have shown that $B^c \subseteq A^c$.

[$\Leftarrow$]. Suppose that $B^c \subseteq A^c$. We want to show that $A \subseteq B$. So suppose that $x \in A$. Then $x \notin A^c$. Since $B^c \subseteq A^c$, again by using the contrapositive of the definition of "$\subseteq$", we see that $x \notin B^c$. Therefore, $x \in B$ and so we have shown that $A \subseteq B$. □

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2. The sets $A \cap B$ and $A \backslash B$ are disjoint.

   Solution. This statement is true.

   Proof. Suppose for contradiction that $A \cap B$ an $A \backslash B$ are not disjoint. Then there exists an element $x \in A \cap B$ and $x \in A \backslash B$. Since $x \in A \backslash B$, $x \in A$ and $x \notin B$. Since $x \in A \cap B$, $x \in A$ and $x \in B$. But then we have that $x \in B$ and $x \notin B$, which is a contradiction. □

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3. If $A \subseteq B$ and $C \subseteq D$ and A and C are disjoint, then B and D are disjoint.

   Solution. This statement is false. For a counterexample, let $A = \{1\}$, $B = \{1, 3\}$, $C = \{2\}$, and $D = \{2, 3\}$. Then clearly $A \subseteq B$ and $C \subseteq D$ and $A \cap C = \{1\} \cap \{2\} = \varnothing$. However, $B \cap D = \{2\} \cap \{2, 3\} = \{2\} \neq \varnothing$. □

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4. If $A \subseteq B$ and $C \subseteq D$ and B and D are disjoint, then A and C are disjoint.

   Solution. This statement is true.

   Proof. Suppose that $A \subseteq B$, $C \subseteq D$, and B and D are disjoint. Suppose for contradiction that A and C are not disjoint so that there exists some $x \in A \cap C$. Then $x \in A$ and $x \in B$. Since $A \subseteq B$ and $x \in A$, therefore $x \in B$. Similarly, since $C \subseteq D$ and $x \in C$, therefore $x \in D$. But now $x \in B$ and $x \in D$, so B and D are not disjoint, which is a contradiction. □

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5. If $A \cap C \subseteq B \cap C$, then $A \subseteq B$.

   Solution. This statement is false. For a counterexample, let $A = \{1, 2\}$ and $B = C = \{1\}$.

   Then $A \cap C = \{1\} = B \cap C$, so the hypothesis holds. However, $A \nsubseteq B$ since $2 \in A$ but $2 \notin B$. □

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6. If $A \cup C \subseteq B \cup C$, then $A \subseteq B$.

   Solution. This statement is false. For a counterexample, let $A = \{1, 2\}$, $B = \{1\}$, and $C = \{1, 2\}$.

   Then $A \cup C = \{1, 2\} = B \cup C$, so the hypothesis holds. However, $A \nsubseteq B$ since $2 \in A$ but $2 \notin B$. □

Problem 4. Let A, B, and C be sets. Prove $A \times (B \backslash C) = (A \times B) \backslash (A \times C)$

Proof. We will prove both containments.

[$\subseteq$]. Suppose $(x, y) \in A \times (B \backslash C)$. Then $x \in A$ and $y \in B \backslash C$. Therefore, $y \in B$ and $y \notin C$. Since $x \in A$ and $y \in B$, therefore $(x, y) \in A \times B$. Since $y \notin C$, we have that $(x, y) \notin A \times C$.

Hence, $(x, y) \in (A \times B) \backslash (A \times C)$. Therefore, we have shown $A \times (B \backslash C) \subseteq (A \times B) \backslash (A \times C)$.

[$\supseteq$]. Suppose $(x, y) \in (A \times B) \backslash (A \times C)$. Then $(x, y) \in A \times B$ and $(x, y) \notin A \times C$. Since $(x, y) \in A \times B$, we have that $x \in A$ and $y \in B$. Since $(x, y) \notin A \times C$ and $x \in A$, this means $y \notin C$. Therefore, $y \in B \backslash C$. Finally, since $x \in A$ and $y \in B \backslash C$, we have that $(x, y) \in A \times (B \backslash C)$.

Therefore, we have shown $A \times (B \backslash C) \supseteq (A \times B) \backslash (A \times C)$. □

Problem 5. Let A and B be nonempty sets. Prove that $A \times B = B \times A$ if and only if $A = B$.

Proof. [⇒] Suppose that $A \times B = B \times A$. We prove that $A \subseteq B$ and $B \subseteq A$. To prove $A \subseteq B$, take an arbitrary $a \in A$. Since B is non-empty, we have some $b_0 \in B$. Then $(a, b_0) \in A \times B = B \times A$, so $(a, b_0) \in B \times A$, which says in particular the $a \in B$, as desired. Conversely, to prove $B \subseteq A$, take an arbitrary $b \in B$ and one fixed $a_0 \in A$, which exists because A is non-empty. Then $(a_0, b) \in A \times B = B \times A$, so $(a_0, b) \in B \times A$, which says in particular that $b \in A$, as desired.

[Alternative for ⇒] We prove this by contrapositive. Suppose that $A \neq B$. This means that either there exists $a \in A$ such that $a \notin B$ or there exists $b \in B$ such that $b \notin A$. Since this problem is symmetric in A and B, without loss of generality, we may assume that we are in the first case.

Choose any element $b \in B$ (which exists since B is nonempty). Then $(a, b) \in A \times B$, since $a \in A$ and $b \in B$. However, $(a, b) \notin B \times A$, since $a \notin B$. Hence, we have shown that $A \times B \neq B \times A$.

[⇐] Suppose $A = B$. By reflexivity, $A \times A = A \times A$. Now by two uses of the substitution axiom (which holds for any kind of equality, not just numbers), we have that $A \times B = B \times A$.

□

Problem 6. For each of the following family of sets $\mathcal{A}$, calculate $\bigcup_{A \in \mathcal{A}} A$ and $\bigcap_{A \in \mathcal{A}} A$. Assume that the universe is $\mathbb{R}$. *You must show that your answer is correct by showing that the given set and your set are equal.*

1. $\mathcal{A} = \{A_z \mid z \in \mathbb{Z}\}$ where $A_z = \left(\frac{2z-1}{2}, \frac{2z+3}{2}\right)$ (Note that this denotes the open interval, not the ordered pair, because the universe is $\mathbb{R}$)

   Solution. Before answering this, we note that

   $$A_z = \left(\frac{2z-1}{2}, \frac{2z+3}{2}\right) = (z - 1/2, z + 3/2)\,.$$

   Claim 1: $\bigcap_{z \in \mathbb{Z}} A_z = \varnothing$.

   Proof of Claim 1: Let $x \in \bigcap_{z \in \mathbb{Z}} A_z$. Thus $x \in A_z$ for **all** $z \in \mathbb{Z}$. In other words, $x \in (z - 1/2, z + 3/2)$ for all $z \in \mathbb{Z}$, i.e., $z - 1/2 < x < z + 3/2$ for all $z \in \mathbb{Z}$. But this is impossible—to see why consider $z = 0$ and $z = 2$. This says that $-1/2 < x < 3/2$ and $3/2 < x < 9/2$, which is a contradiction. Thus x cannot exist, so $\bigcap_{z \in \mathbb{Z}} A_z = \varnothing$.

   Claim 2: $\bigcup_{z \in \mathbb{Z}} A_z = \mathbb{R}$.

   Proof of Claim 2: Since $\mathbb{R}$ is the universe, $A_z \subseteq \mathbb{R}$ automatically, so $\bigcup_{z \in \mathbb{Z}} A_z \subseteq \mathbb{R}$ for free. We will now show that the other inclusion holds. Let $x \in \mathbb{R}$ arbitrary. Then there is some integer n such that $n \leqslant x \leqslant n + 1$[1]. Notice that $[n, n + 1] \subseteq A_n$, thus $x \in A_n$. Therefore $x \in \bigcup_{z \in \mathbb{Z}} A_z$ by the definition of union. Thus $\bigcup_{z \in \mathbb{Z}} A_z = \mathbb{R}$. This completes our proof. □

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[1] We don't really now why this is true. It can be proven using the Well-Ordering Principle and the so-called Archimedean Property. I will accept this claim with no proof.

2. $\mathcal{A} = \{A_n \mid n \in \mathbb{Z}_{>0}\}$ where $A_n = \left[0, \frac{1}{n}\right]$.

   Solution. Claim 1: $\bigcap_{n\in\mathbb{Z}_{>0}} A_n = \{0\}$.

   Proof of Claim 1: Certainly $0 \in A_n$ for all $n \in \mathbb{Z}^+$, so it's clear that $\{0\} \subseteq \bigcap_{n\in\mathbb{Z}^+} A_n$. We will show the other set inclusion by *contrapositive*: Let $x \in \mathbb{R}$ such that $x \neq 0$. If $x < 0$, then $x \notin A_1$ so $x \notin \bigcap_{n\in\mathbb{Z}^+} A_n$. If instead $x > 0$, then there exists a natural number m such that $m > 1/x$ (This is exactly the Archimedean Property or $\mathbb{R}$, but I will allow this claim with no proof). Thus $x > 1/m$, so $x \notin A_m$. Therefore $x \notin \bigcap_{n\in\mathbb{Z}^+} A_n$. Thus $\bigcap_{n\in\mathbb{Z}^+} A_n \subseteq \{0\}$. This completes our proof.

   Claim 2: $\bigcup_{n\in\mathbb{Z}^+} A_n = [0, 1]$

   Proof of Claim 2: Notice that $A_1 = [0, 1]$, which means that $[0, 1] \subseteq \bigcup_{n\in\mathbb{Z}^+} A_n$.

   For the reverse inclusion, assume that $x \in \bigcup_{n\in\mathbb{Z}^+} A_n$. Thus $x \in A_n$ for some $n \in \mathbb{Z}^+$. Therefore $0 \leqslant x \leqslant 1/n \leqslant 1$, so $x \in [0, 1]$. This proves our claim. □

Problem 7. Let Λ be a nonempty index set and let $\mathcal{A} = \{A_\alpha \mid \alpha \in \Lambda\}$ be an indexed family of sets.

1. Let B be a set such that $B \subseteq A_\alpha$ for all $\alpha \in \Lambda$. Show that $B \subseteq \bigcap_{\alpha\in\Lambda} A_\alpha$.

   Proof. (If $B = \varnothing$ then this is automatically true, so assume $B \neq \varnothing$.) Let $x \in B$. Since $B \subseteq A_\alpha$ for all $\alpha \in \Lambda$, $x \in A_\alpha$ for all $\alpha \in \Lambda$. By definition of intersection, thus $x \in \bigcap_{\alpha\in\Lambda} A_\alpha$. This shows that $B \subseteq \bigcap_{\alpha\in\Lambda} A_\alpha$. □

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2. Assume instead that $A_\alpha \subseteq B$ for all $\alpha \in \Lambda$. What can we conclude in this case? (Prove that your answer is correct.)

   Solution. The strongest statement we can prove in this case is that $\bigcup_{\alpha\in\Lambda} A_\alpha \subseteq B$. Here's the proof:

   Let $x \in \bigcup_{\alpha\in\Lambda} A_\alpha$. Thus $x \in A_\alpha$ for some $\alpha \in \Lambda$. Since $A_\alpha \subseteq B$, thus $x \in B$. This shows that $\bigcup_{\alpha\in\Lambda} A_\alpha \subseteq B$. □

Problem 8. Complete Exercise 17 on page 198 of the textbook (*Mathematical Reasoning* by Sundstrom).

Proposition (a). For each natural number n with $n \geqslant 2$, $2^n > 1 + n$.
Critique: The original write-up did not show the base case! However, the result still holds. Here's a better proof.

Proof. We will show that if $n \in \mathbb{N}$ with $n \geqslant 2$, then $2^n > 1 + n$.
Base case. Our base case is $n = 2$. Notice that

$$2^n = 2^2 = 4 > 3 = 1 + 2 = 1 + n.$$

Thus $2^n > 1 + n$ when $n = 2$.

Induction step. Let k be a natural number with $k \geqslant 2$ and assume that

$$2^k > 1 + k. \tag{1}$$

We will show that $2^{k+1} = 1 + (k+1)$. Observe that

$$\begin{aligned} 2^{k+1} = 2 \cdot 2^k &> 2(1+k) \quad \text{by Equation (1)} \\ &= 2 + 2k \\ &> 2 + k \quad \text{because } k > 0 = 1 + (k+1). \end{aligned}$$

This completes our induction step.

Therefore the proposition is proved by induction. □

Proposition (b). Each natural number greater than or equal to 6 can be written as the sum of natural numbers, each of which is a 2 or a 5.

Critique: Notice that $P(10)$ is *not* covered in the base case or the induction step! This is a huge flaw, and the proof as it stands is worthless! Here's an ironclad proof, and it uses 2 in place of 5, which makes our lives easier.

Proof. Let $n \in \mathbb{N}$ with $n \geqslant 6$. We will show that there exist $a, b \in \mathbb{Z}_{\geqslant 0}$ such that $n = 2a + 5b$. We will prove this by induction.

Base case. Observe that $6 = 2 \cdot 3 + 5 \cdot 0$ and $7 = 2 \cdot 1 + 5 \cdot 1$. So the above statement holds for $n = 6, 7$.

Induction step. Assume that the statement holds for some $k \in \mathbb{N}$ with $k \geqslant 7$. We will show that it holds for $k+1$. Notice that $k - 1 \geqslant 6$, so the statement holds for $k-1$ either by the case case (if $k = 7, 8$) or the induction hypothesis (if k is larger). Thus $k - 1 = 2a + 5b$ for some $a, b \in \mathbb{Z}_{\geqslant 0}$. Therefore

$$\begin{aligned} k + 1 &= (k-1) + 2 \\ &= (2a + 5b) + 2 \\ &= 2(a+1) + 5b. \end{aligned}$$

Thus the statement holds for $k+1$ too. This completes our proof by induction. □

Problem 9. Do not turn in, but complete the progress checks in the sections of the textbook.